

LU Decomposition

* To factorize $A = LU$

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 3 & 6 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Standard LU Decomp without row swap:

$$E_1: R_2 \leftarrow R_2 - 3/2 R_1 \Rightarrow E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -3/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 3/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_1 A = A_1: \begin{bmatrix} 2 & 2 & 1 \\ 0 & 3 & -1/2 \\ 1 & 1 & 1 \end{bmatrix}$$

$$E_2: R_3 \leftarrow R_3 - \frac{1}{2} R_1 \Rightarrow E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{1}{2} & 0 & 1 \end{bmatrix}$$

$$E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & 1 \end{bmatrix}$$

$$E_2 E_1 A = A_2 = \begin{bmatrix} 2 & 2 & 1 \\ 0 & 3 & -\frac{1}{2} \\ 0 & 0 & \frac{1}{2} \end{bmatrix} = U.$$

$$E_2^{-1} E_1^{-1} U \stackrel{?}{=} A \Rightarrow \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ \frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}} \begin{bmatrix} 2 & 2 & 1 \\ 0 & 3 & -\frac{1}{2} \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ \frac{3}{2} & 1 & 0 \\ \frac{1}{2} & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1/2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 2 & 1 \\ 3 & 6 & 1 \\ 0 & 0 & 1/2 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 2 & 1 \\ 3 & 6 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

② When you begin with Row swap, P matrix does the row swap, also called the PERMUTATION matrix.

$$PA = LU$$

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 3 & 6 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$R_3 \leftrightarrow R_1 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} A.$$

$$PA = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 6 & 1 \\ 2 & 2 & 1 \end{bmatrix}$$

$$E_1: R_2 \leftarrow R_2 - 3R_1$$

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_1 A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & -2 \\ 2 & 2 & 1 \end{bmatrix}$$

$$E_2: R_3 \leftarrow R_3 - 2R_1$$

$$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \quad E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

$$E_2(E_1 A) = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & -2 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\Rightarrow U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & -2 \\ 0 & 0 & -1 \end{bmatrix}$$

$$E_2 E_1 (PA) = U.$$

$$\Rightarrow PA = E_1^{-1} E_2^{-1} U$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & -2 \\ 0 & 0 & -1 \end{bmatrix}$$

$$= \begin{matrix} L & U \\ \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} & \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & -2 \\ 0 & 0 & -1 \end{bmatrix} \end{matrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 3 & 6 & 1 \\ 2 & 2 & 1 \end{bmatrix} = PA.$$